TRAINING-FREE · NO TASK HEAD · OPEN VOCABULARY

Test-time compute of jina-embeddings-v5-omni for image tagging

THIS DECK

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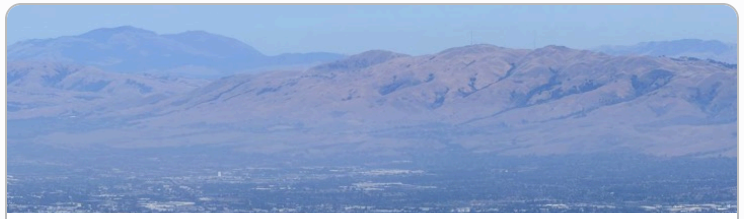
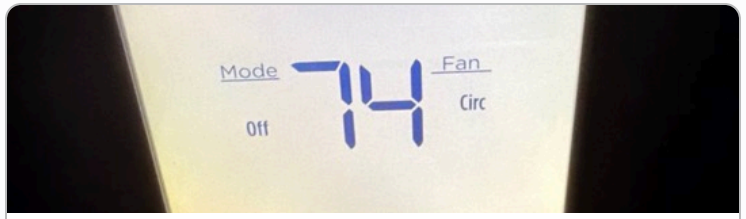
THE RESULT, FIRST

Training-free, open-vocabulary tagging from a frozen embedding model.



mountain · phoenix · sky · rooftop · location



clock · thermostat · dial · dst · wifi

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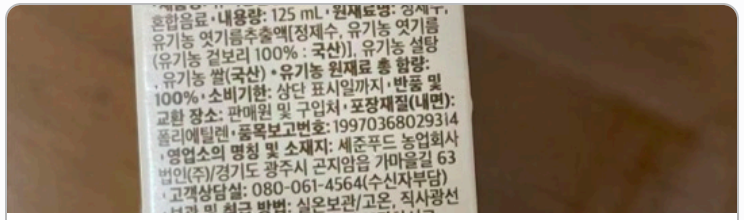
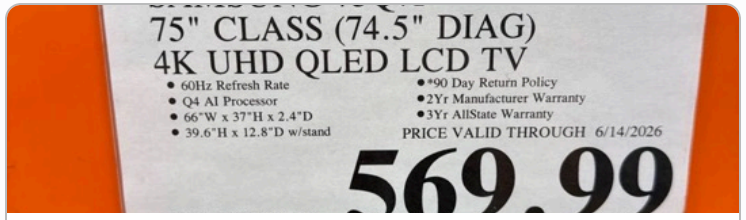
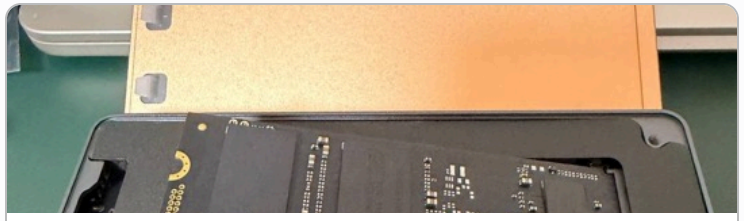


contiene · liters · dosage · thirst · consumo

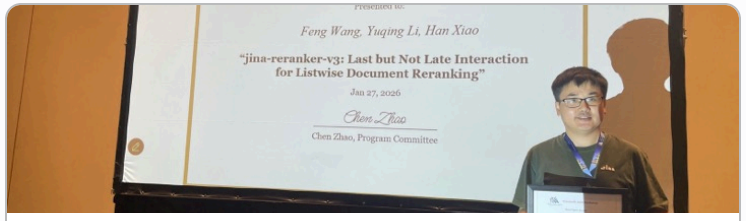


harga · lcd · samsung · refurbished · warrants

sunglasses · watches · branded · bronze · necklace



tablet · motherboard · wearable · celular · datastore



award · forwarding · speaking · defendant · workshop

gift · toy · celebration · balloons · cheerful

SETUP

Problem formulation.

GIVEN	one embedding model, **jina-embeddings-v5-omni-nano** (~1B, image and text in one 768-d space), and one image.
OUTPUT	words for **every** object in the image: multi-label, open vocabulary.
FROZEN	all weights. The model is a retrieval encoder; it has no tagging head, no classifier, and was never trained for this task.
ALLOWED	inference-time computation over the model's own outputs, and nothing else.

NO TRAINING · NO SECOND MODEL · NO WORDNET / DICTIONARY · NO REGEX / POS TAGGER

Whatever tagging ability appears must come from test-time compute of the embedding model.

A TAXONOMY

How test-time scaling works on frozen embedding models.

A frozen encoder's inference-time budget takes roughly three forms:

A DEEPER PASS

Read more of the one pass you already ran: every patch vector, not just the pooled one. The retrieval world's **late interaction**.

new information: reclaimed from inside the pass

B MORE PASSES

Run the frozen encoder again on new views: split a document, crop an image, re-encode.

new information: acquired from the input

C BOOTSTRAP

Transform the vectors you already have: whitening, propagation, optimal transport. Where most of the training-free literature lives.

new information: none

Same frozen encoder in all three; they differ only in what enters the computation.

RESEARCH QUESTIONS

Two research questions.

RQ1 **Can a frozen embedding model become an image tagger through test-time compute alone?**

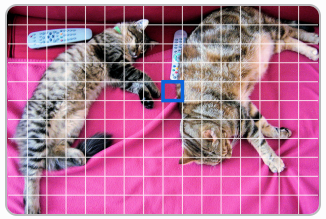
Prior tagging work pays with training or resources: Tag2Text / RAM / RAM++ train over ~6,400 curated tags; TagCLIP (AAAI'24) and PIAA ("[CLS] is not enough") are training-free but assume a given class list, on two-tower CLIP plus WordNet or POS tools.

RQ2 **If it can: which test-time compute scales its accuracy, bootstrapping or new information?**

The 2024–26 literature bets on bootstrapping: whitening (PIAA), label propagation (ZLaP), optimal transport (OTTER), Bayesian adaptation (BCA).

THE MODEL · ARCHITECTURE

Image and text embedding space.



PER IMAGE · TEST TIME

cat

coche

櫥

× 128,260 tokens

VOCABULARY · ENCODED ONCE

jina-embeddings-v5-omni-nano

Qwen3-VL vision tower

16 px patches
2×2 merge

EuroBERT-12L text tower

bidirectional · 768-d

vision tokens injected at image-token positions

never modified, never retrained

P: one 768-d row per patch (the tower's internal 16 px unit), each vector shaped by attention over the *whole* image

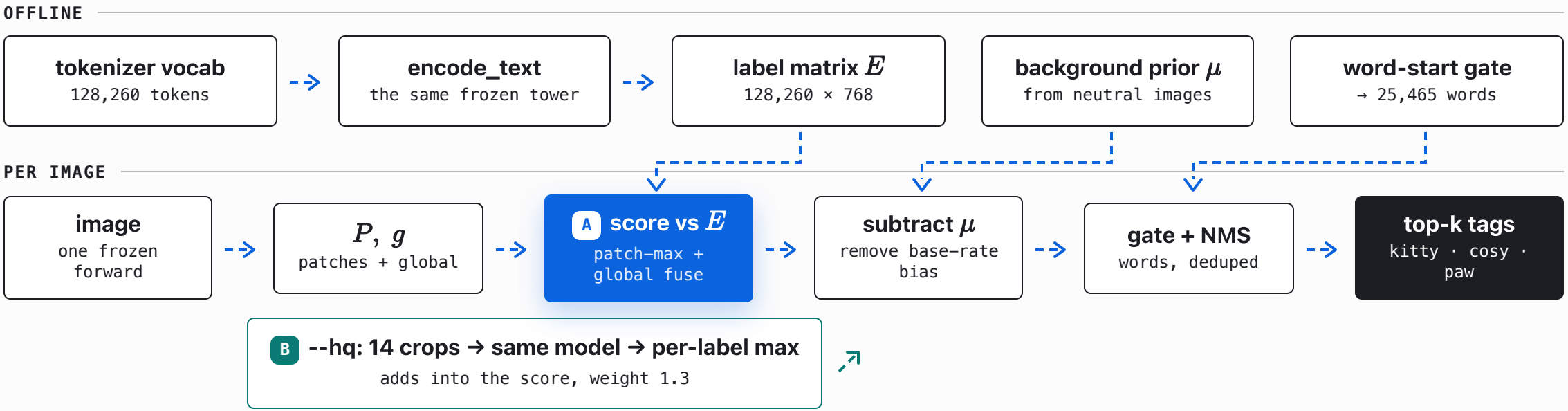
g: one vector for the whole image, pooled from the last token of the same sequence

E: label matrix, 128,260 × 768; gated to 25,465 words at scoring

Both paths end in the same space, so $\langle p,\ e_\ell \rangle$ is well-defined: any patch of any image, scored against any word the model knows.

THE PIPELINE

The complete tagger workflow as test-time compute.



Every box is the frozen model or plain arithmetic: A reads more of one pass; B runs new passes on new pixels.

STEP 1 · THE LABEL SPACE

The label set is the tokenizer's own vocabulary.

No curated tag list. Take all **128,260** tokens and encode each one as text into a label matrix

$$E = \left[e_\ell \right]_{\ell \in V}, \qquad e_\ell = \mathbf{encode\_text}(\ell) \in \mathbb{R}^{768}, \quad |V| = 128{,}260$$

One matmul now scores any image against the entire vocabulary; fragments are filtered later by a word-start gate (step 4).

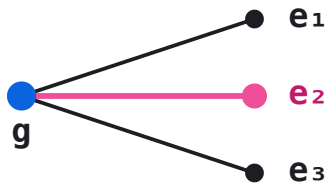
```
wrong: image_vec · embed_tokens.weight
       → Tutor, avatar, PyTuple  (garbage)

right: image_vec · encode_text(token)
      → kitty, cosy, plush, paw  (aligned)
```

Why encode_text, not the embedding table? The model has `tie_word_embeddings = false`: the input-embedding table lives in a different space than the pooled output. You must encode each token *as a text string* to put labels in the image's space.

STEP 2 · WHERE TO LOOK

Score every patch.

GLOBAL POOLED

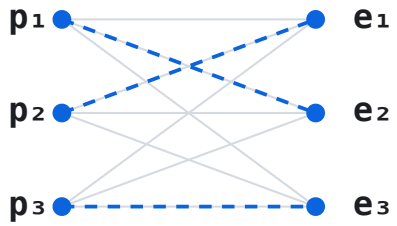


$$s_\ell = \langle g, e_\ell \rangle$$

one vector dominated by the most salient object

mAP 0.264

PER-LABEL MAX OVER PATCHES



$$s_\ell = 0.7 \max_{p \in P} \langle p, e_\ell \rangle + 0.3 \langle g, e_\ell \rangle$$

each label keeps its strongest patch match — late interaction, ColBERT-style

mAP 0.635

A The patch vectors already exist in the forward pass. Scoring each label against all of them, and keeping the max, is one extra step over outputs you already have.

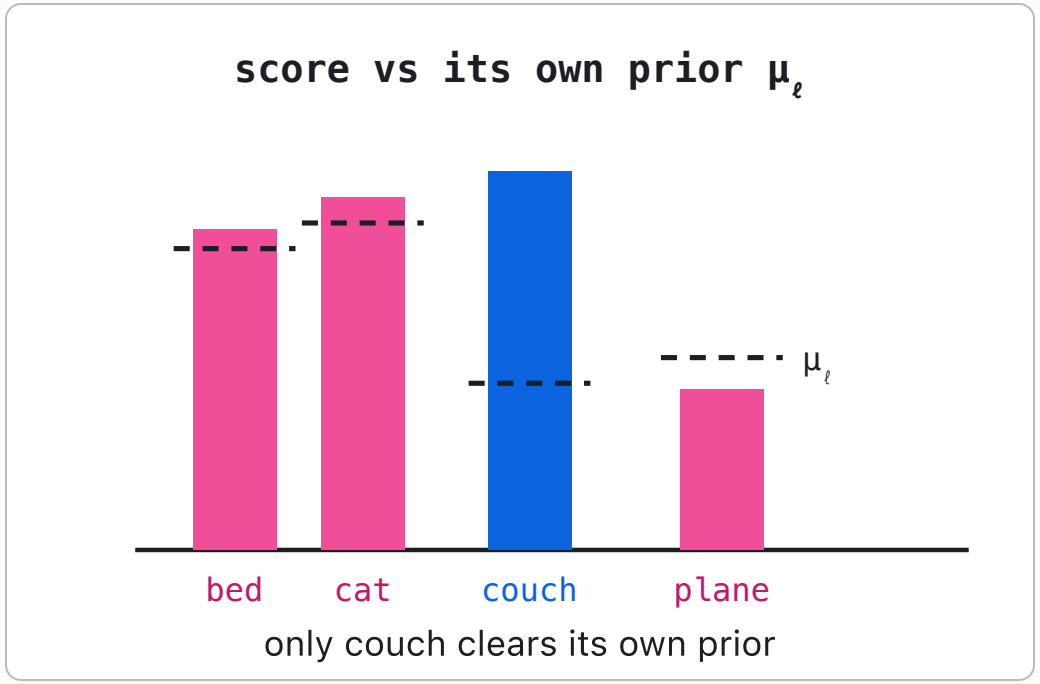
STEP 3 · DE-BIAS

Subtract the per-label prior.

Raw cosine has a base-rate bias: generic words ("bed", "cat") score high on almost any image because they sit near the image modality. Estimate a per-label prior once from neutral background images, then subtract it:

$$\tilde{s}_\ell \;=\; s_\ell - \mu_\ell, \qquad \mu_\ell = \mathbb{E}_{x \sim \text{background}}\big[s_\ell(x)\big]$$

No calibration set, no labels, one offline pass.

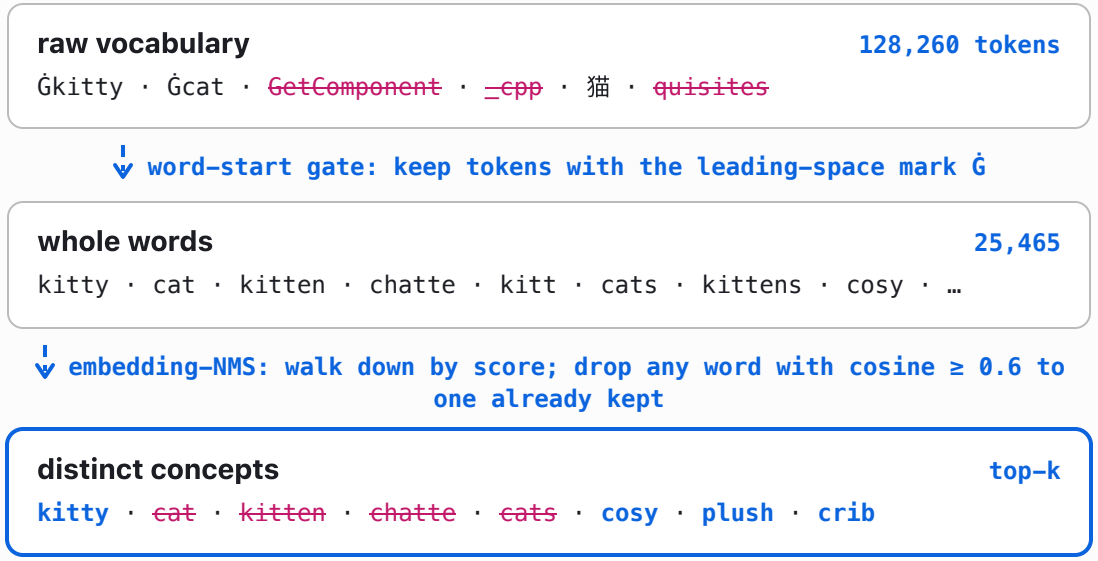


STEP 4 · CLEAN UP

The tokenizer already knows what a word is.

Word-start gate. Byte-BPE tokenizers encode a leading space as the glyph Ġ; a token beginning with it starts a new word: `Ġcat` is a whole word, `GetComponent` is a fragment. That one built-in signal is the gate; no external dictionary.

Embedding-NMS. Synonyms and multilingual variants (/ Cat / кот / cat) collapse into one, using cosine in the model's own space. Walk labels by score; keep ℓ only if $\max_{k \in \text{kept}} \langle e_\ell, e_k \rangle < 0.6$. Again: no lookup table, just the geometry.



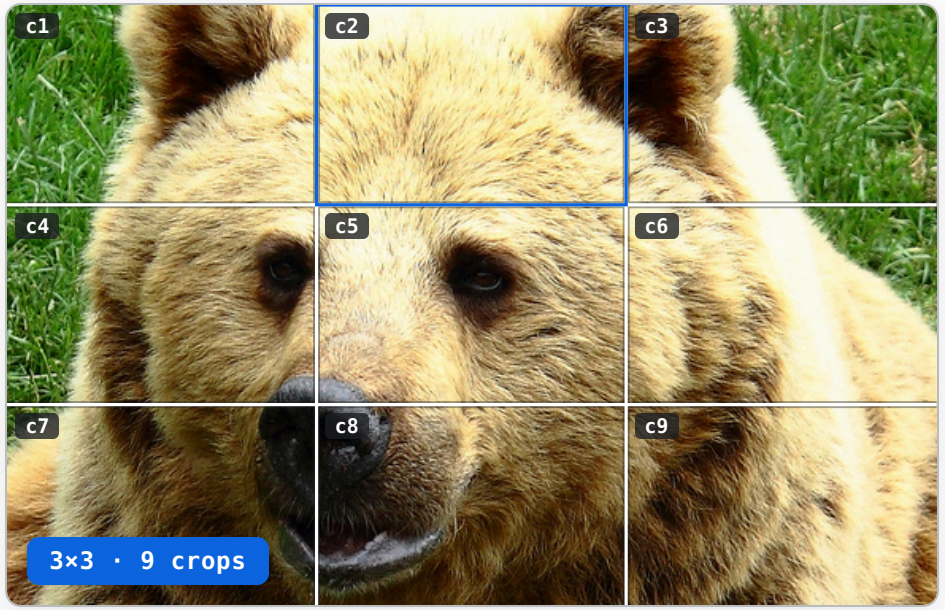
STEP 5 · NEW VIEWS

Multi-crop re-encoding.

Class-Wise Reranking (CWR), the crop-and-rescore idea from TagCLIP (AAAI'24), extended to a full-coverage grid of 14 crops (overlaid right) through the same frozen model. A crop is a *new input image*, itself re-patchified by the tower; a small object fills whichever crop contains it, so weak signal becomes strong.

$$S_\ell = \tilde{s}_\ell + 1.3\Big(\max_{c=1..14} s_{c,\ell} - \mu_\ell\Big), \qquad s_{c,\ell} = \max\Big(\max_{p\in P_c}\langle p, e_\ell\rangle,\ \langle g_c, e_\ell\rangle\Big)$$

B Per-label **max**, not averaging: averaging hurt. The outlier crop is the evidence.



fast: wolf, roar, muzzle → --hq: fur, **bear**, muzzle
multi-crop re-encoding corrects the single-pass misreading

ALL FIVE STEPS, ASSEMBLED

The entire tagger is one scoring function.

the same score , applied to each crop as a fresh image

$$S(\ell) = \underbrace{0.7\left(\max_{p \in P}\langle p,\, e_\ell\rangle - \mu_\ell\right)}_{\textbf{patch evidence}} + \underbrace{0.3\left(\langle g,\, e_\ell\rangle - \mu_\ell^{g}\right)}_{\textbf{global context}} + \underbrace{1.3\left(\max_{c=1..14} s_\ell(\text{crop}_c) - \mu_\ell\right)}_{\textbf{multi-crop re-encode}}$$

A **Patch evidence**: algebra over rows the forward pass already produced. Free. +0.37 mAP.

A **Global context**: the pooled vector, de-biased by its own background prior. Free.

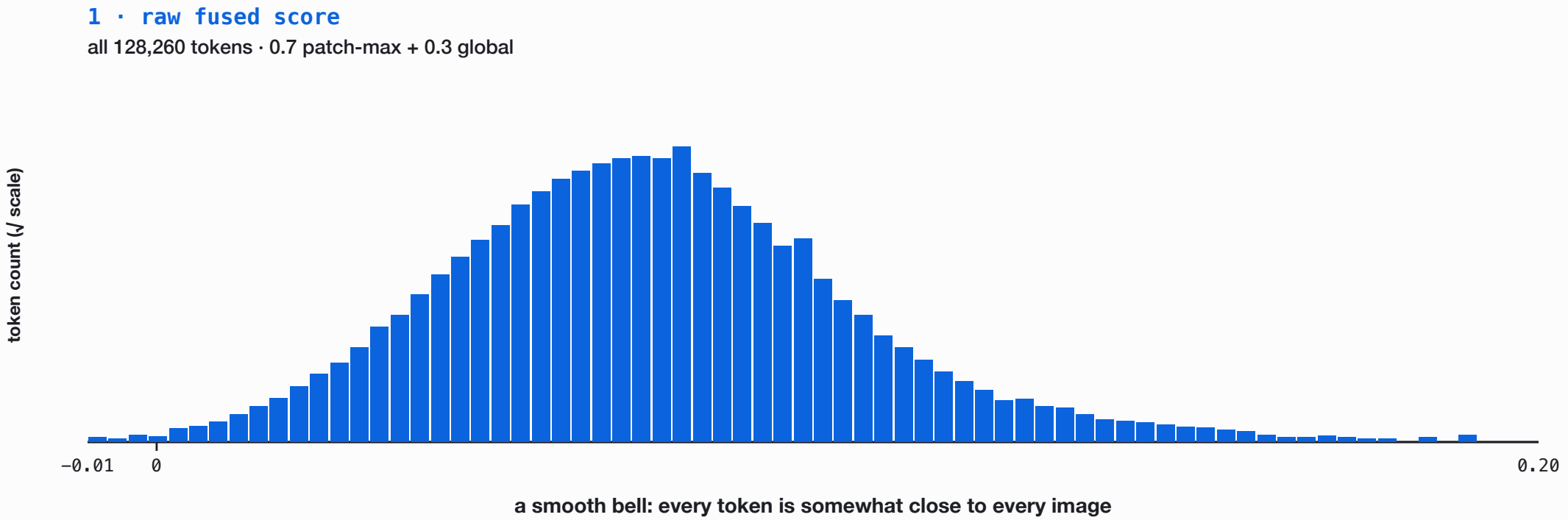
B **Multi-crop**: 14 fresh forward passes on new pixels. 14× the cost. +0.075 mAP.

then $\ell \in \textbf{word-gated vocabulary} \rightarrow \textbf{embedding-NMS}\ (\tau = 0.6) \rightarrow \textbf{top-}k$

No head, no logits, no learned threshold: three fixed weights, two background priors, and max operations over frozen-model outputs.

THE EQUATION, VISUALIZED

The pipeline sharpens a distribution.



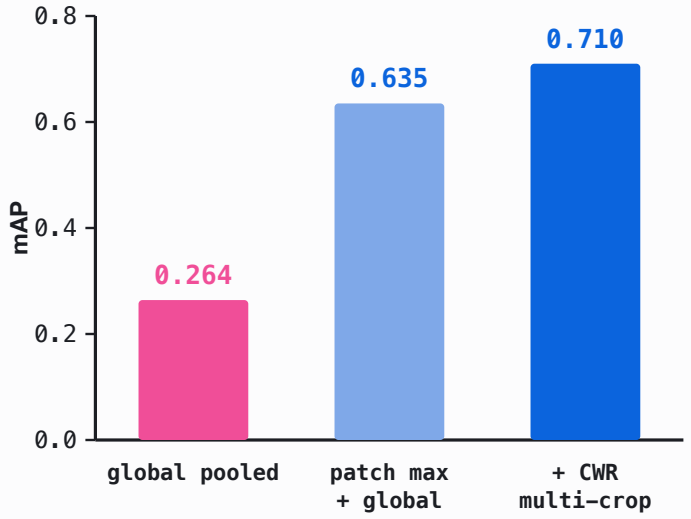
Real pipeline output, one image, all 128,260 tokens. Tagging is distribution sharpening.

RESULTS

Results on COCO-150.

METHOD	P@1	P@3	R@5	MAP
global pooled	0.433	0.289	0.449	0.264
patch max + global	0.753	0.427	0.631	0.635
softpool	0.773	0.449	0.649	0.608
+ CWR multi-crop	0.813	0.476	0.680	0.710

150 COCO-val images, real multi-label ground truth, avg 2.93 labels/image. softpool trades mAP for top-k precision; the ladder tracks the max-pool path.

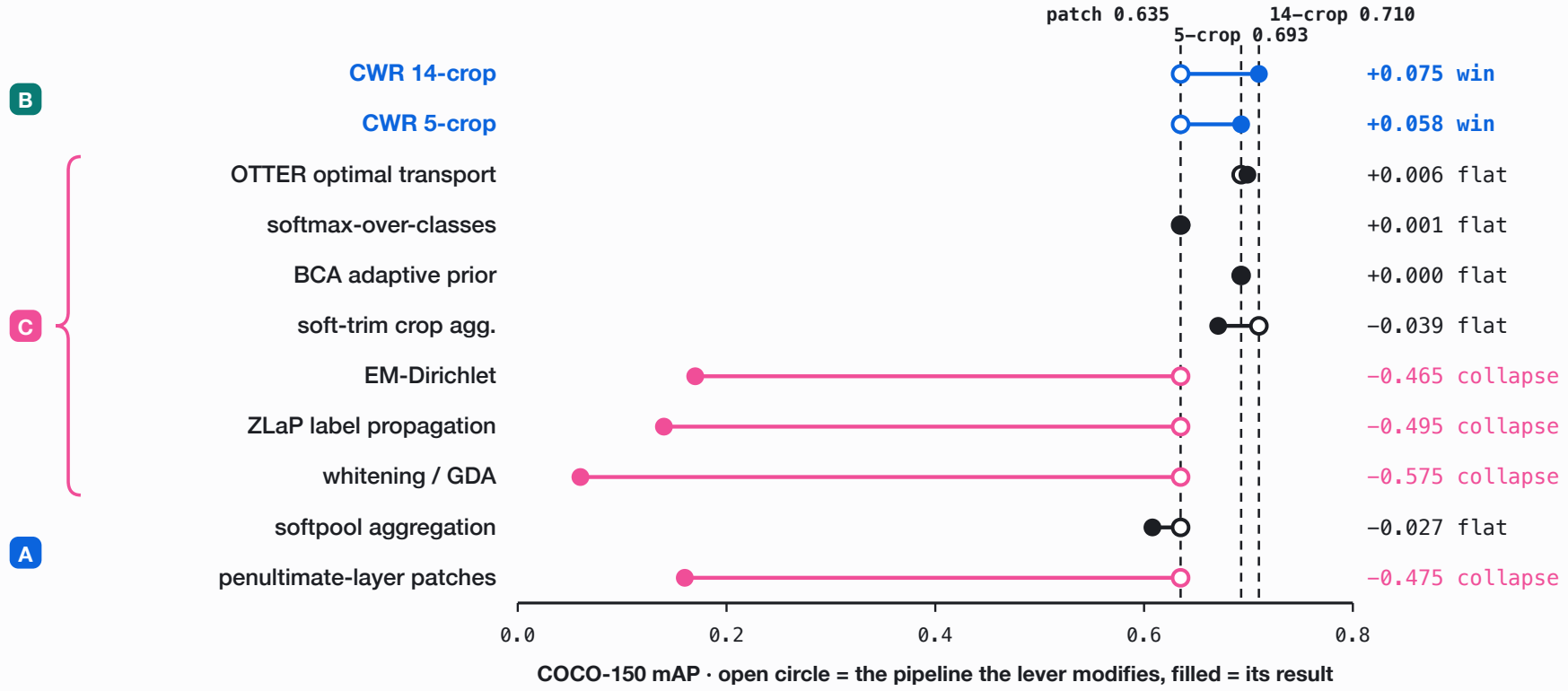


mAP: global → patch → +CWR

THE EXPERIMENT

Does bootstrapping scale accuracy at test time?

One example: **OTTER** re-balances scores toward a target class distribution, which only shuffles vectors it already has. Its open circle sits at the 5-crop baseline of **0.693**, and it lands at 0.699.



THE META-CONCLUSION

Only new information moves accuracy.

Bootstrap methods were designed for weakly calibrated, single-label, two-tower CLIP. This encoder's space is already aligned, calibrated, and multi-label: **there is nothing left to recover**.

C BOOTSTRAP

whitening 0.06 · ZLaP 0.14 · EM-Dirichlet 0.17 · OTTER +0.006 over its own base. None improves the pipeline it modifies.

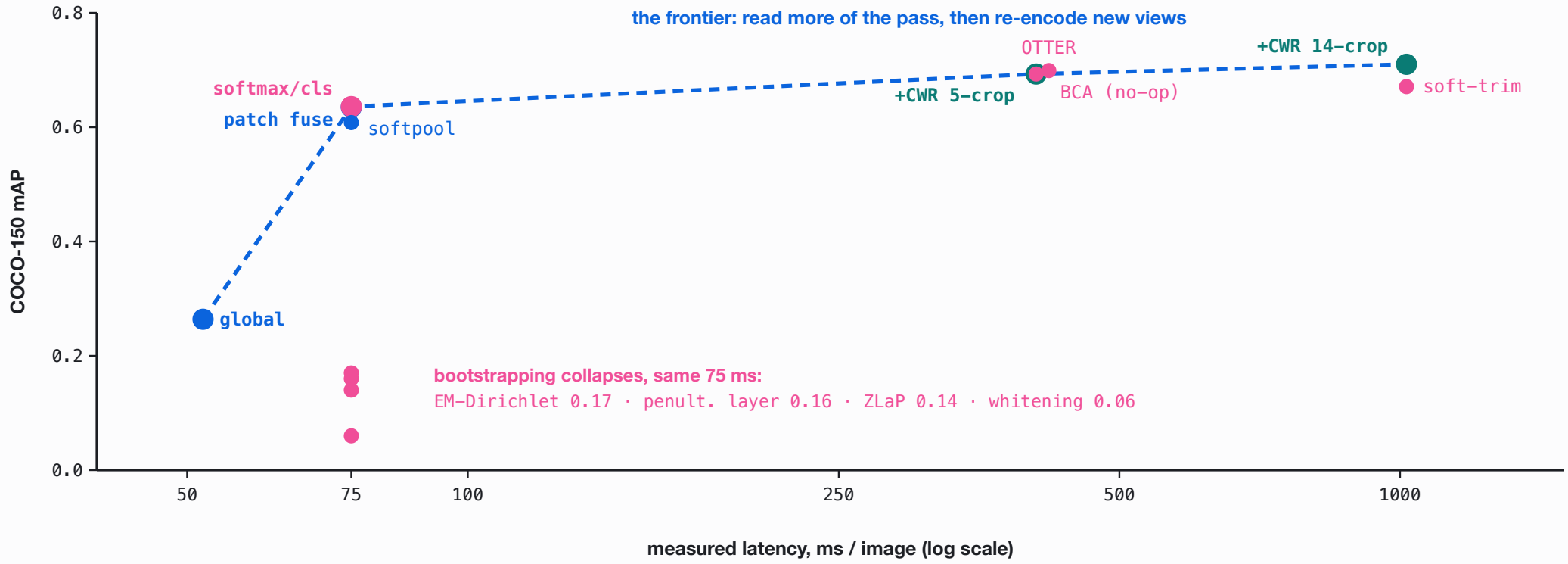
B MORE PASSES

CWR crops → mAP 0.710, P@1 0.813. The only intervention that improves results.

The ceiling is the 1B model itself. Real test-time compute here means giving the model more to look at, not re-arranging what it already saw.

THE FRONTIER

Accuracy versus test-time compute.



The frontier is traced by reading more of the pass and re-encoding new views; bootstrapping sits at the same cost, at or below it (<0.1 ms/img, measured).

ONE MORE MODE · --NGRAM

Extend to n-grams.

Every tag so far is a single word. Can more **test-time compute** turn it into a grounded **noun phrase**, with no part-of-speech tagger and no grammar?

→

ttc tagging pipeline

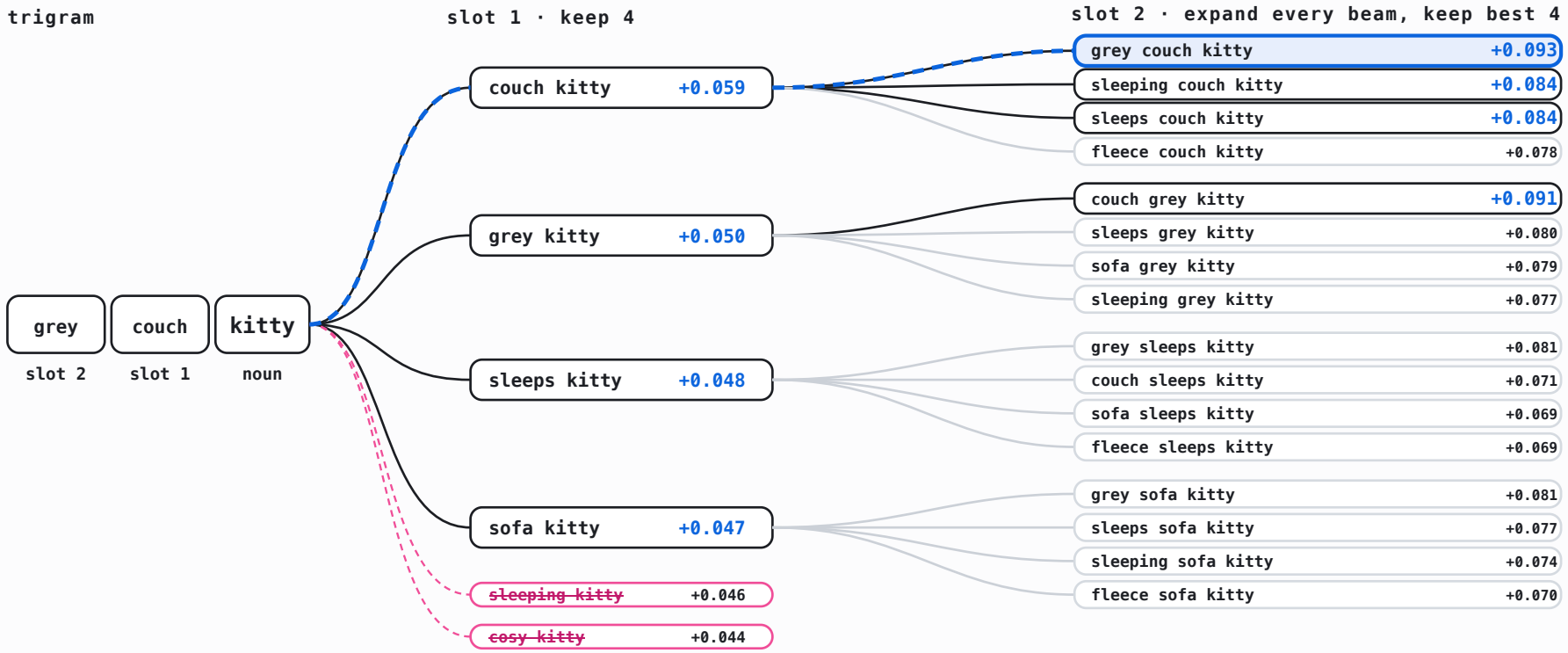
`--ngram beam search`

→

`n = 1`	`kitty` `cosy` `plush` `crib`
`n = 2`	`couch kitty` `black cosy` `grey plush` `grey crib`
`n = 3`	`grey couch kitty` `sleeping black cosy` `grey sleeping plush` `sleeps grey crib`

Beam search over n-gram slots.

Each slot is filled one at a time; the encoder scores the full phrase against the region: $\langle E(\text{phrase}), \text{region} \rangle - \langle E(\text{kitty}), \text{region} \rangle$. Beam width 4.



The winning path re-orders its own words: “grey couch kitty” beats “couch grey kitty”. Word order, resolved by an embedding model with no grammar.

QUALITATIVE RESULTS

n-gram tagging in action.

DEFAULT	onstage · attendees · livest · concert · presenter
CROPS	onstage · venue · ceiling · projector · theater
N=2	conference onstage · projector attendees · attendees livest · projector concert · projector presenter
N=3	conference crowd onstage · lobby projector attendees · projector attendees livest · projector lobby concert · attendees projector presenter

DEFAULT	driv · rims · ford · traction · sidew
CROPS	carro · rims · windshield · steering · beach
N=2	windshield driv · windshield rims · sail ford · windshield traction · blue sidew
N=3	blackjack windshield driv · dealership windshield rims · windshield yacht ford · yacht sedan traction · blue car sidew

DEFAULT	attendees · passengers · protestors · migrants · crowded
CROPS	attendees · commuters · demonstrators · synagogue · crowded
N=2	protester attendees · demonstrators passengers · passengers protestors · attendees migrants · demonstrators crowded
N=3	passengers protester attendees · crowded demonstrators passengers · passenger attendees protestors · passengers protester migrants · passenger demonstrators crowded

Default, crop re-encoding, beam-searched n-grams. Top-5 per mode, unstaged photos. Verbatim, unfiltered.

RELATION TO PRIOR WORK

Versus the recent tagging literature.

LINE OF WORK	THEIR SETUP	OUR DELTA
Tag2Text / RAM / RAM++	trained tagging models, curated ~6.4k tag list	training-free, label space = tokenizer vocab
TagCLIP	CLIP two-tower, penultimate layer, DMAR+CWR	single omni model, last layer is the output space
PIAA	patch-level scoring (“[CLS] is not enough”) + GDA whitening	patch thesis confirmed (+0.37 mAP); whitening collapses (0.06)
recent calibration (OTTER, BCA)	calibration set or target prior to de-bias	one offline background prior, zero calibration

Both questions come out the way we hoped. A frozen embedding model does become a tagger with no training, and the only compute that scales it is the kind that adds new information.

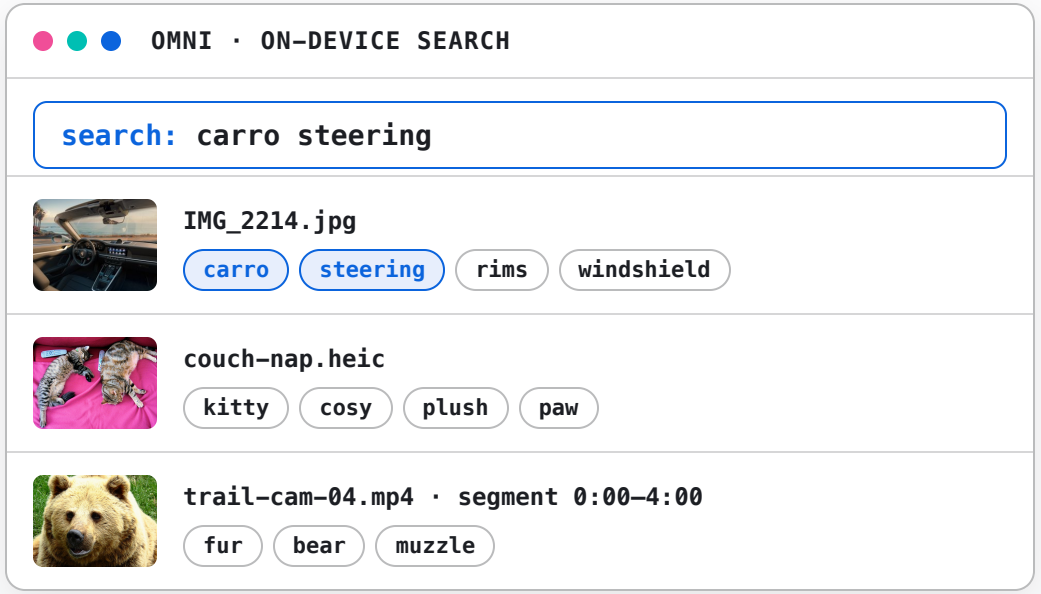
DEPLOYMENT

On-device deployment in Omni.

OMNI-MACOS

The Python study became a feature of **Omni**, a native on-device semantic-search app (Swift + MLX, same frozen model). The port reuses the architecture directly:

SAME FORWARD PASS	the patch rows already exist for the file's *embedding*; tagging adds one matmul.
~0.6 MS / IMAGE	about 4% overhead on the embed step. Tagging is effectively free at index time.
TAGS = SNIPPETS	tags land in the search snippet: media becomes *keyword*-searchable.
SELF-CALIBRATING	the background prior μ is estimated on-device from the first 64 images it sees, then frozen.



tag chips as they appear in results; the third row is a video segment

ONE RULE, EVERY MEDIA SHAPE

Images, video, and scanned documents.

IMAGE

Tagged at **index time**, inside the embedding forward. Patch-max + global, prior-centered, gated, NMS-deduped.

HQ IMAGE

A background **re-tag pass** adds the production CWR variant: 5 crops, per-label max.

VIDEO

32 frames per **240-second segment**, one sequence through the tower: patch-max pools over **space and time**.

SCANNED PDF

Pages render to images, same path: **per-page tags** ("invoice, table, signature") as the snippet.

THE BIGGER PICTURE

Test-time compute for embedding models.

A frozen embedding model can do more than its training reveals. You reach the rest by spending compute at inference, not by adding parameters. Both of my talks push on that same idea.

AIE SF TALK

Spend test-time compute to get more **relevance**, better retrieval on the task the encoder was already trained for.

THIS TALK

Spend test-time compute to get a new **capability**, image tagging, a task the encoder was never trained for.

Scaling test-time compute of an embedding model unlocks tasks beyond retrieval.

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